

Reconfigurable Battery Management System for Rovers

Engineering Review Document — B.Tech Capstone Project

Document Purpose	Detailed technical design for peer review and feedback
Project Scope	BMS with string-level reconfiguration (rover is test platform only)
Pack Configuration	6× Li-Ion 21700 cells in 3S2P nominal
Switching Granularity	String-level (not individual cell-level)
Core Innovation	SOC/SOA-driven string disconnect, fault isolation, and charge optimization
Document Date	July 2026

1. Problem Statement

Traditional battery management systems for small vehicles (rovers, drones, ROVs) treat the battery pack as a fixed electrical configuration. Cells are permanently wired in series or a fixed series-parallel arrangement, and the BMS acts purely as a passive watchdog — it monitors parameters and disconnects the entire pack when any single cell exceeds safe limits. This architecture has three fundamental limitations.

1.1 Binary Fault Response

When any cell develops a fault (thermal anomaly, voltage irregularity, resistance spike), the traditional BMS disconnects the entire pack. For a rover mid-mission, this means total power loss. There is no mechanism to isolate a single problematic cell or string while continuing on healthy cells. One cell's failure becomes the entire system's failure.

1.2 Stranded Energy Due to Cell Imbalance

In a fixed series string, discharge terminates when the weakest cell hits undervoltage cutoff, even if remaining cells still hold 30–50% charge. Published research on LG M50T 21700 cells shows that cell-to-cell variation starts at 0.45% in capacity and 1.68% in internal resistance even for brand-new cells from the same batch. After 287 cycles, internal resistance increases by approximately 25%, and this increase is non-uniform across cells. Gogoana et al. (MIT, 2013) demonstrated that a 20% difference in internal resistance between parallel-connected cells reduces pack cycle life by 40%.

1.3 No Charging Optimization

In a fixed 3S2P configuration, charging behavior is dictated by the pack topology. Cells cannot be independently equalized during charging without external balancing circuits (passive resistors that waste energy as heat, or active balancing circuits that add cost and board space). A Vitesco and CEA (2024) study on a self-reconfigurable battery demonstrator showed 22% faster charging compared to conventional packs when cells could be reconfigured to all-parallel during charging.

2. Proposed Solution — What We Are Building

A battery management system for a 6-cell Li-ion pack that can electrically disconnect and reconnect battery strings in real time using MOSFET switches. The BMS monitors every cell individually (voltage, temperature, SOC) and makes string-level switching decisions based on SOC and SOA conditions. The rover serves only as a test platform to generate realistic load profiles; the project is the BMS, not the vehicle.

Critical design clarification: This system operates at string level, not individual cell level. You cannot bypass one cell in a 3S string and parallel the remaining 2S (7.4V) with a healthy 3S (11.1V) string — the voltage mismatch would drive destructive current. When a cell faults, the entire string containing it is disconnected. This is an intentional, physics-driven design constraint, not a limitation to be solved with a DC-DC converter.

2.1 Pack Architecture

Parameter	Specification
Cell count	6 individual cells
Cell type	21700 Li-ion (e.g., Samsung 50E: 3.7V, 5Ah, 10A continuous)
Nominal configuration	3S2P — two parallel strings of 3 series cells
Nominal voltage	11.1V (range: 7.5V fully depleted to 12.6V fully charged)
Nominal capacity	10Ah (55.5 Wh per string, 111 Wh total)
String A	Cell 1 + Cell 2 + Cell 3 in series
String B	Cell 4 + Cell 5 + Cell 6 in series

Table 1: Battery pack specifications

2.2 Operating Modes

The BMS has four distinct operating modes. Each mode has a clear entry condition and switching sequence.

Mode	Configuration	When Active	What Happens
Normal Operation	3S2P — both strings paralleled	Default mode; both strings healthy and SOC matched	Both strings share load equally via natural current splitting. Simplest, most efficient mode.
Alternation Mode	One string active, one string resting; swap every 2–3 min	Optional mode for wear leveling demonstration	Bulk capacitor holds bus voltage during 1–2ms switch gap. Buck converter keeps MCU alive through sag.
Fault Isolation	One string disconnected; rover runs on remaining string	Any cell in a string exceeds SOA boundary	Faulty string fully disconnected. Rover continues at 11.1V with half capacity (5Ah).
Charging Mode	All 6 cells in parallel	Pack plugged in for charging	All cells equalize voltage naturally. Precharge sequence used before parallel connection.

Table 2: Operating modes

3. Detailed Operation

3.1 Normal Operation (Both Strings Paralleled)

In normal mode, both strings are connected in parallel to the output bus. Current splits naturally between the two strings based on their internal resistance (Ohm's law). If cells are well-matched, each string carries approximately half the total load current. Both strings deplete together, and the full 10Ah capacity is available. The BMS continuously monitors per-cell voltage and temperature but does not actively switch anything. This is the baseline mode and the most efficient — no switching overhead, no transients, no capacitor needed.

Research from Vitesco and CEA (2024) showed only 6% energy consumption improvement from active reconfiguration during normal driving on WLTP cycles. For a healthy, well-matched pack under normal load, reconfiguration provides marginal benefit. The real value emerges in fault, imbalance, and charging scenarios.

3.2 Alternation Mode (Wear Leveling Demonstration)

In this optional mode, the BMS alternates which string powers the rover. String A runs for 2–3 minutes, then the BMS switches to String B. This equalizes wear across both strings, preventing the resistance mismatch that Gogoana et al. showed causes 40% lifetime reduction.

The switching sequence: the BMS opens the active string's MOSFET and closes the resting string's MOSFET. During the 1–2ms transition gap, a bulk capacitor (4700µF, 25V) on the output bus holds the voltage. The bus may sag from 11.1V to approximately 9V for 2ms. The rover's buck converter (which steps pack voltage down to 3.3V/5V for the MCU) accepts an input range of 4.5–40V, so a sag to 9V is well within range. The MCU never loses power. Motors are unaffected by a 2ms dip.

Why include this if the benefit is marginal at 2 strings? The MOSFETs to disconnect each string exist anyway for fault isolation (mandatory feature). Alternation is purely a firmware feature using hardware that's already present — zero additional component cost. At 2 strings the wear leveling benefit is 5–10%. But this is a proof-of-concept for larger systems: Cârstoiu et al. (2021) demonstrated 415% improvement in operating time with wear leveling on large-scale reconfigurable packs with many strings.

3.3 Fault Isolation

When any cell in a string crosses an SOA boundary, the entire string containing that cell is disconnected from the bus. The rover continues operating on the remaining healthy string at 11.1V with 5Ah capacity (half the nominal). A traditional BMS would shut down the entire pack — this is the project's strongest, most defensible claim.

Fault Condition	Detection Method	Response
Cell overtemperature (>50°C or rising >2°C/min)	NTC thermistor on each cell, read via MCU ADC every 100ms	Disconnect entire string containing hot cell. Continue on healthy string.
Cell voltage sag under load (V < 3.0V while SOC > 20%)	Per-cell voltage monitoring via ADC or BQ76952	Indicates high internal resistance (degraded cell). Disconnect string, flag for inspection.
Cell undervoltage (V < 2.5V)	Per-cell voltage monitoring	Cell is depleted. Disconnect string to prevent deep discharge damage.

Cell overvoltage (V > 4.2V during charging)	Per-cell voltage monitoring	Stop charging that string. If in parallel charging mode, isolate affected string.
Overcurrent (string current > cell rating)	INA226 current sensor per string	If both strings active: load is too high, alert vehicle controller. If one string: reduce load.

Table 3: Fault conditions and responses

3.4 Charging Mode

When the pack is plugged in, the BMS reconfigures all 6 cells into parallel. In parallel, current flows naturally from higher-voltage cells to lower-voltage cells until all equalize. This provides automatic voltage balancing without any external balancing hardware. Charging current is shared across all 6 cells equally, enabling the fastest possible charge rate.

3.5 Precharge Sequence (Critical Safety Feature)

Before any parallel connection is established — whether reconnecting strings in normal mode, entering charging mode, or reconnecting after alternation — the BMS executes a precharge sequence. This prevents the destructive current spike that would occur if voltage-mismatched sources are directly paralleled.

The problem without precharge: If String A is at 10.5V and String B is at 11.7V, direct paralleling through MOSFETs with ~50–70mΩ total path resistance causes an instantaneous current spike of $1.2V / 0.07\Omega = 17A$. This can blow MOSFETs, damage cells, or trigger thermal runaway.

The precharge circuit: A separate, small-signal MOSFET connects the two strings through a 10Ω resistor. When activated, the initial current is only $1.2V / 10\Omega = 0.12A$. This trickle current transfers energy from the higher-voltage string to the lower-voltage string (not wasted — the lower string gets charged). After 100–200ms, the voltage difference drops below 0.1V. The main parallel MOSFET then closes safely (spike = $0.1V / 0.07\Omega = 1.4A$, completely harmless). The precharge MOSFET opens. The resistor is no longer in the circuit. Total energy lost in the resistor: approximately 0.07 Wh (0.0006% of pack energy).

4. What This BMS Does NOT Do

These are intentional design boundaries, not limitations to be patched. Each was evaluated and rejected for specific engineering reasons.

Rejected Feature	Why Rejected
Individual cell bypass within a series string	Removing one cell from a 3S string creates a 2S string at 7.4V. This cannot be paralleled with the healthy 3S string at 11.1V. The voltage mismatch is structural, not a balancing problem.
DC-DC converter to normalize output voltage	If a converter can regulate any voltage to what the load needs, reconfiguration becomes pointless. The converter alone would solve the problem, making the switching hardware redundant.
Per-cell MOSFET switching (18 MOSFETs)	String-level switching achieves fault isolation and charging optimization with far fewer components. Per-cell switching adds cost and complexity without proportional benefit at 6-cell scale.
Machine learning for reconfiguration decisions	Rule-based logic with fixed thresholds is deterministic, verifiable, and safe. ML introduces unpredictability in a power management system where wrong decisions cause hardware damage.
Voltage mode switching (series for cruise, parallel for power)	The load (motor driver) expects stable ~11.1V. Switching between 11.1V series and 3.7V parallel would require a DC-DC converter (see above) or damage the motor driver.

Table 4: Intentionally excluded features with rationale

5. System Architecture

5.1 Hardware Components

Subsystem	Component	Qty	Function
Battery	21700 Li-ion cells (Samsung 50E or similar)	6	Energy storage; 3S2P nominal config
String Switching	N-MOSFET (high-side or low-side per string)	TBD (4–6)	Connect/disconnect each string from bus
Precharge	Small N-MOSFET + 10Ω 2W resistor	1 set	Safe parallel reconnection
Bus Capacitor	4700μF 25V electrolytic	1	Hold bus voltage during alternation switch gap
Voltage Sensing	Per-cell voltage (divider + ADC or BQ76952)	6 channels	Cell-level voltage monitoring
Current Sensing	INA226 with shunt resistor	2	Per-string current measurement
Temperature	NTC 10K thermistor	6	Per-cell temperature monitoring
MCU	STM32F446 or ESP32	1	Control, algorithm, communication

Power Regulation	Buck converter (LM2596 / MP1584)	1	Pack voltage to 3.3V/5V for electronics
Comm	CAN transceiver or USB-UART	1	Data logging and vehicle interface
Safety	Fuse + kill switch	1 each	Hardware emergency disconnect

Table 5: Hardware component list

5.2 Monitoring vs. Switching Granularity

A key architectural decision: monitoring is per-cell but switching is per-string. Every cell's voltage, temperature, and derived SOC is tracked individually. This is essential because degradation data (25% resistance increase after 287 cycles, non-uniform across cells) shows that cell-level visibility is necessary to make informed string-level decisions. However, the MOSFETs that control power flow operate at the string level — connecting or disconnecting entire strings of 3 cells, not individual cells.

This mirrors industrial practice: Tesla monitors every cell individually but manages power at the module level. The BQ76952 battery monitor IC (used in Paper 4's AUV BMS) can monitor up to 16 cells on a single chip via multiplexed ADC, making per-cell monitoring straightforward without per-cell switching hardware.

6. Switching Architecture Detail

6.1 String Connection Circuit

Each string connects to the output bus through a main MOSFET (or MOSFET pair for bidirectional blocking). Additionally, a single precharge path exists between the two strings consisting of a small MOSFET in series with a 10Ω resistor. The total MOSFET count is significantly fewer than the 18 required for per-cell switching — estimated at 4–6 main switching MOSFETs plus 1 precharge MOSFET.

6.2 Switching Sequences

Operation	Sequence	Duration
String swap (alternation)	1. Open active string MOSFET 2. Capacitor holds bus (1–2ms) 3. Close resting string MOSFET 4. Bus stabilizes	~2–5ms total
Parallel reconnection	1. Close precharge MOSFET (current via 10Ω) 2. Monitor voltage difference 3. Wait until $\Delta V < 0.1V$ (~100–200ms) 4. Close main parallel MOSFET 5. Open precharge MOSFET	~100–300ms total
Fault isolation	1. Open all MOSFETs on faulty string 2. Verify disconnection (V reading) 3. Remaining string continues	~1–2ms immediate
Enter charging mode	1. Disconnect from load (open load MOSFET) 2. Precharge sequence for cell paralleling 3. Connect charger 4. All cells charge in parallel	~300ms for precharge setup
Exit charging mode	1. Disconnect charger 2. Reconfigure to 3S2P 3. Verify string voltages match 4. Connect to load	~100ms

Table 6: Switching sequences for each operation

6.3 Anti-Oscillation Protection

A minimum cooldown timer of 500ms is enforced between any topology changes. This prevents rapid oscillation between configurations if a trigger condition is borderline (e.g., temperature hovering around the 50°C threshold). Without this, the system could switch every 100ms control cycle, creating continuous transients.

7. Sensing and State Estimation

7.1 Measurements

Measurement	Sensor	Resolution	Update Rate
Cell voltage (x6)	ADC via voltage divider or BQ76952	~1mV	100ms
String current (x2)	INA226 + shunt resistor	~10mA	100ms
Cell temperature (x6)	NTC 10K thermistor via ADC	~0.5°C	1 second

7.2 SOC Estimation

Per-cell SOC is estimated using an Extended Kalman Filter (EKF) with coulomb counting as the prediction step and open-circuit voltage as the correction step. The EKF runs on the MCU every 100ms. SOC feeds two decisions: (1) when to disconnect a depleted string (any cell SOC < 20%), and (2) wear leveling scheduling in alternation mode (prioritize the higher-SOC string for active duty).

7.3 SOA Monitoring

The Safe Operating Area is defined by three boundaries per cell: voltage (2.5V–4.2V), current (continuous discharge rating from cell datasheet), and temperature (0–60°C for discharge, 10–45°C for charging). Any cell crossing any SOA boundary triggers immediate string disconnection. SOA triggers have higher priority than SOC-based decisions.

8. Edge Cases and Failure Modes

Scenario	System Response
Both strings develop faults simultaneously	Controlled full-pack shutdown with warning. This is the only scenario where the system behaves identically to a traditional BMS.
All cells below 20% SOC	Low-power alert sent to vehicle controller. Rover should begin return-to-base. BMS does not force shutdown until 2.5V cutoff.
MOSFET fails stuck-open	Detected by voltage sensing (string shows no current flow when it should). String permanently excluded from operation.
MOSFET fails stuck-closed	Detected by unexpected current path. Adjacent switches opened for isolation. Emergency shutdown if isolation fails.
Rapid load transients (terrain bumps, direction changes)	No switching triggered. These are normal load events handled by the paralleled strings. Only sustained overcurrent triggers action.
Voltage mismatch detected before parallel reconnection	Precharge sequence activates automatically. Parallel connection only after $\Delta V < 0.1V$. If equalization fails after 5 seconds, abort and flag error.
Charger connected while strings are at different SOC	Precharge sequence equalizes before full parallel. Then all cells charge together. Most imbalanced cell gets slightly more current naturally.
Ambient temperature below 0°C	Under-temperature protection prevents charging (lithium plating risk). Discharge allowed at reduced current only.

Table 8: Edge cases and failure mode handling

9. Quantitative Claims and Supporting Evidence

Every claim in this project is backed by published data. No unsupported performance numbers.

Claim	Value	Source
Fault tolerance: traditional vs ours	0% output vs 50% output when one string faults	Fundamental: KCL in series circuit makes this binary
Energy stranded by cell imbalance	10–15% of pack energy wasted per cycle at 10% SOC drift	Tang et al. 2026, Section 4.2: series cutoff equation
Resistance mismatch impact on lifetime	20% resistance mismatch → 40% lifetime reduction	Gogoana et al. 2013 (MIT), Journal of Power Sources
Current imbalance from impedance mismatch	30% impedance difference → 60% peak current difference	Bruen and Marco, experimental study

Wear leveling benefit (large scale)	Up to 415% operating time improvement	Cârstoiu et al. 2021, Int. J. Energy Research
Charging speed improvement	22% faster than conventional pack	Vitesco + CEA 2024, SRB demonstrator (WLTP cycles)
Normal operation efficiency gain	Only ~6% energy consumption reduction	Vitesco + CEA 2024 (honest: marginal during normal use)
BMS power overhead	~0.5–0.6W continuous	Calculated from component datasheets
New cell variation (beginning of life)	0.45% capacity, 1.68% resistance CoV	LG M50T 21700 study (2024), 40 cells tested
Resistance increase after cycling	~25% after 287 cycles	Pouch cell aging study (2021)

Table 9: All quantitative claims with sources

10. Honest Limitations

Limitation	Impact	Mitigation
Cannot bypass individual cells within a string	One bad cell takes out an entire 3-cell string (50% capacity lost)	Acceptable trade-off: string-level switching is simpler and avoids voltage mismatch problems
Marginal benefit during normal operation of a healthy, matched pack	Only ~6% improvement when everything is working perfectly	Value is in fault and degradation scenarios, not normal operation. Honestly stated in report.
Wear leveling benefit is small at 2 strings	5–10% estimated; 415% requires many strings	This is a POC. Same algorithm scales to larger packs where benefit is substantial.
MOSFET conduction losses add overhead	~300mW at 5A through 2–3 MOSFETs in path	Select ultra-low RDS(on) MOSFETs. Loss is <1% of pack output power.
Switching transients during alternation	1–2ms power gap; ~2V bus sag	Bulk capacitor holds bus. Buck converter rides through sag. MCU unaffected.
6-cell pack too small for dramatic results	Limited reconfiguration options compared to 100+ cell pack	POC validates concept. Future scope: module-level reconfiguration for larger systems.

Table 10: Limitations honestly assessed

11. Test and Validation Plan

11.1 Milestone 0 — Simulation

Build a MATLAB/Simulink model of the 6-cell pack using first-order RC (Thevenin) equivalent circuit per cell. Simulate three scenarios with fixed topology vs reconfigurable: normal operation (baseline comparison), one cell fault at 50% SOC (runtime comparison), and 15% SOC imbalance between strings (energy extraction comparison). Deliverable: comparison plots proving concept before any hardware investment.

11.2 Milestone 1 — Hardware Verification

Fabricate the switching PCB. Connect to 6 cells on bench with resistive load (not rover). Manually trigger each switching operation via test firmware. Verify: MOSFETs switch cleanly, precharge sequence works, voltage and current readings are accurate, no shoot-through. Deliverable: bench test log with oscilloscope captures of switching transients.

11.3 Milestone 2 — Integrated Testing

Mount BMS on rover. Run controlled comparison: same rover, same route, same pre-conditioned battery with deliberate 15% SOC imbalance. Run 1: BMS locked in fixed 3S2P (simulating traditional BMS). Run 2: BMS fully active. Compare total runtime, energy extracted, and demonstrate fault isolation by heating one cell mid-mission with a heat gun. Deliverable: logged data, comparison charts, live demo.

12. Estimated Bill of Materials

Category	Item	Est. Cost (■)
Battery	6x 21700 cells + holders	2,100
Switching	4–6x power MOSFETs + 1x precharge MOSFET	800–1,200
Switching	Gate driver ICs (if needed)	400–600
Passive	1x precharge resistor (10Ω 2W)	10
Passive	1x bulk capacitor (4700μF 25V)	50
Sensing	2x INA226 current sensor	300
Sensing	6x NTC 10K thermistor	120
Sensing	1x BQ76952 (optional, replaces voltage dividers)	800–1,200
Control	1x STM32 or ESP32 dev board	800
Power	1x buck converter module (LM2596)	100
Comm	1x CAN transceiver or USB-UART	150
PCB	Custom PCB (2oz copper, 2-layer)	500–1,000
Safety	Fuse + kill switch + connectors (XT60/XT30)	300
Rover	Chassis + motors + motor driver + RC controller	2,000–3,000
Misc	Wire, solder, heatsinks, enclosure	500
	TOTAL ESTIMATE	~9,000–12,000

Table 11: Bill of materials estimate

13. Future Scope

Direction	Description	Expected Impact
Module-level reconfiguration	Replace per-cell monitoring with per-module monitoring. Apply switching at module level for 48V+ systems.	Enables scaling to EV and home energy storage. Same algorithm, different granularity.
More strings (3S3P, 3S4P)	Adding more parallel strings increases reconfiguration options and fault tolerance.	Losing one string from three = 67% capacity vs current 50%.
Second-life battery support	Apply to packs assembled from retired cells with high parameter variation.	Maximum benefit scenario: heterogeneous cells need reconfiguration most.
ML-based optimization	Add reinforcement learning layer on top of rule-based engine for mission-specific strategy tuning.	Potentially better wear leveling schedules. Keep rule-based as safety fallback.

Table 12: Future development directions

14. References

- [1] F. Tang, J. Ma et al., "Reconfigurable Battery System Topology, Operation and Feature: A Comprehensive Survey," Journal of Energy Storage, vol. 168, 122573, Elsevier, 2026.
- [2] A. Harakare et al., "Design of Battery Management System for an Autonomous Underwater Vehicle," OCEANS 2022 - Chennai, IEEE, 2022.
- [3] I. Saidulu et al., "A Smart BMS Development for Electric Vehicles to Improve Energy and Power Management Structure," ICSES, IEEE, 2024.
- [4] S. Pushparaj et al., "Improving the Longevity and Efficiency of Electric Vehicle Batteries with Innovative Battery Management Technologies," ICCES, IEEE, 2024.
- [5] R. Gogoana et al., "Internal Resistance Matching for Parallel-Connected Lithium-Ion Battery Cells," Journal of Power Sources, MIT, 2013. (20% mismatch → 40% lifetime reduction)
- [6] T. Bruen, J. Marco, "Modelling and Experimental Evaluation of Parallel Connected Li-ion Cells," Journal of Power Sources. (30% impedance → 60% current imbalance)
- [7] G. Cârstoiu et al., "Novel Battery Wear Leveling Method for Large-Scale Reconfigurable Battery Packs," Int. J. Energy Research, 2021. (415% operating time improvement)
- [8] Vitesco Technologies + CEA, "Simulation and Testing of Self-Reconfigurable Battery Advanced Functions for Automotive Application," 2024. (6% energy, 22% charging improvement)
- [9] N. Bouchhima et al., "Lifetime of Self-Reconfigurable Batteries Compared with Conventional Batteries," Journal of Energy Storage, vol. 15, pp. 400–407, 2018.
- [10] P. Schuster et al., "Lithium-Ion Battery Degradation: Comprehensive Cycle Ageing Data for Commercial 21700 Cells," Journal of Power Sources, 2024. (0.45% capacity, 1.68% resistance CoV)
- [11] S. Ci, N. Lin, D. Wu, "Reconfigurable Battery Techniques and Systems: A Survey," IEEE Access, vol. 4, pp. 1175–1189, 2016.